

E-Commerce Management System

Module: Payment Management Module — End User & Technical Documentation

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GitHub Repo: ansarizeenat/ecommerce-category-management	Live Server (Render): https://ecommerce-category-management.onrender.com
Tech Stack: Java 25, Spring Boot 4.1, JPA, MySQL 8.0, Thymeleaf	Date: September 2026

1. Introduction

The **Payment Management Module** handles payment processing and records transaction history for each order. It integrates with third-party payment gateways (simulated Stripe, PayPal, and bank transfer) to securely process payments. Administrators can view transaction history, retry failed charges, and issue refunds for cancelled or returned orders.

2. Database Design & Specification

Table Name: **payments** — schema matches the Itvedant assignment.

Table 1: payments

Column	Data Type	Key / Constraint	Description
payment_id	INT	PK, Auto-Increment	Unique identifier for each payment
order_id	INT	FK (orders.id)	Foreign key referencing the order
amount	DECIMAL(10,2)	NOT NULL	Amount paid
payment_method	VARCHAR(50)	NOT NULL	Credit Card, Debit Card, PayPal, Bank Transfer
payment_status	VARCHAR(50)	NOT NULL	Paid, Failed, or Refunded
created_at	DATETIME	AUTO-POPULATED	Timestamp when the payment was processed
updated_at	DATETIME	AUTO-UPDATED	Timestamp when payment details were last updated
transaction_id	VARCHAR(100)	OPTIONAL	Simulated Stripe / PayPal / bank gateway reference
refund_reason	VARCHAR(300)	OPTIONAL	Reason recorded when a Paid payment is refunded

3. Core Functionalities

1. Process Payment: Payments are processed through external payment gateways (e.g., Stripe, PayPal). Payment methods include credit/debit cards, PayPal, and bank transfers. Amount defaults to the order total. Duplicate Paid charges and cancelled orders are rejected. Failed gateway attempts are stored as Failed.

2. Payment Dashboard: Admins can view transaction history, including payment ID, amount, payment method, and status. Filters: All, Paid, Failed, Refunded. Summary cards show collected revenue and refund totals.

3. Refund Payment: Admins can issue refunds for cancelled or returned orders. Refund transactions are logged (status Refunded) and displayed on the dashboard. The linked order is cancelled.

4. End User Steps

1. Open <https://ecommerce-category-management.onrender.com> and select the **Payment Management** tab.
2. Place an order in Order Management if no unpaid order exists.
3. Choose the unpaid order and a payment method, then click **Charge Gateway**.
4. Confirm the new row on the dashboard (Paid or Failed).
5. Click **Refund** on a Paid payment for a cancelled or returned order.

5. REST API Endpoint Specifications

Method	Endpoint	Description	Sample Payload / Response
POST	<code>/api/payments</code>	Process payment via simulated gateway	<code>{"orderId":1, "paymentMethod":"Credit Card"}</code>
GET	<code>/api/payments</code>	Transaction history (supports <code>?status=Paid</code>)	JSON list with payment ID, amount, method, status
GET	<code>/api/payments/stats</code>	Dashboard totals	totalRevenue, paidCount, failedCount, totalRefunded
GET	<code>/api/payments/{id}</code>	Get payment by ID	Returns a single payment record
POST	<code>/api/payments/{id}/refund</code>	Refund a Paid payment and cancel the order	<code>{"reason":"Order returned"}</code>

6. Automated Test Suite

Tests in `PaymentServiceTest.java` cover successful charge, duplicate Paid rejection, cancelled-order rejection, refund, double-refund rejection, status filters, stats, and failed-gateway logging.

7. Assignment Submission Summary

Item	Details & Links
1. GitHub Repository	https://github.com/ansarizeenat/ecommerce-category-management
2. Free Cloud Hosted URL	https://ecommerce-category-management.onrender.com
3. End User Documentation	This PDF — Payment Management Module End User Documentation.
4. Modules Completed	Category, Product, Order, Customer, and Payment Management.